Graphene and 2D Materials: Science, Processing, and Applications (ChE 494)

Homework 1: Science of 2D Materials

September 10, 2019 **Due Date: September 17, 2019** **Total: 100**

Submit Your Homework by September 17, 2019 at 4:00 pm in EIB 142 or by E-Mail to sbehura1@uic.edu

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1. Graphene is a two-dimensional (2D) crystal with carbon atoms arranged in a honeycomb structure as shown in Figure 1. Using the above information and Figure 1, answer the following questions.

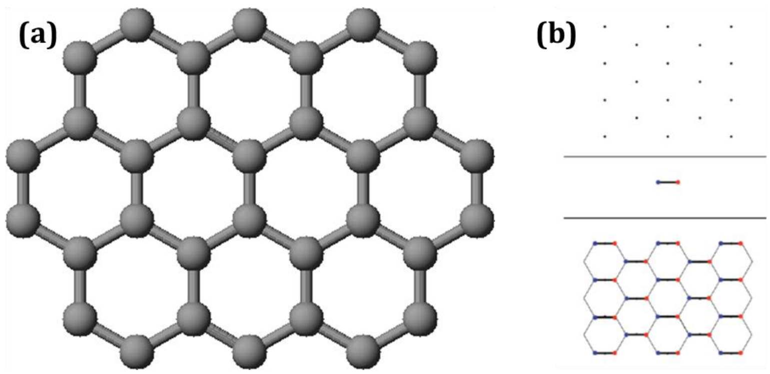

Figure 1: Graphene Lattice.

   (a) Assuming that the unit cell is a hexagon of carbon atoms (Figure 1 (a)), how many atoms of carbon are there in each unit cell (Explain)?
   (b) Crystal Structure = Lattice + Basis. In Figure 1 (b), designate: (i) lattice and its type, (ii) basis and how many atoms in basis, and crystal structure.

2. What is Stone-Wales (SW) defect in graphene and state SW defect's consequences on the properties of graphene?

3. Intrinsic monolayer graphene is a zero bandgap ($E_g = 0$ eV) semiconductor. Show the band structure (Energy-Momentum Linear Dispersion) of an intrinsic monolayer graphene. Suggest some techniques (minimum two) to open bandgap in graphene.

4. Calculate the lattice unit vectors (a_1 and a_2) in the following honeycomb graphene crystal lattice.

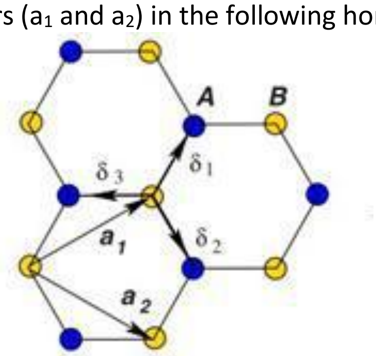

Figure 2: Graphene Lattice.